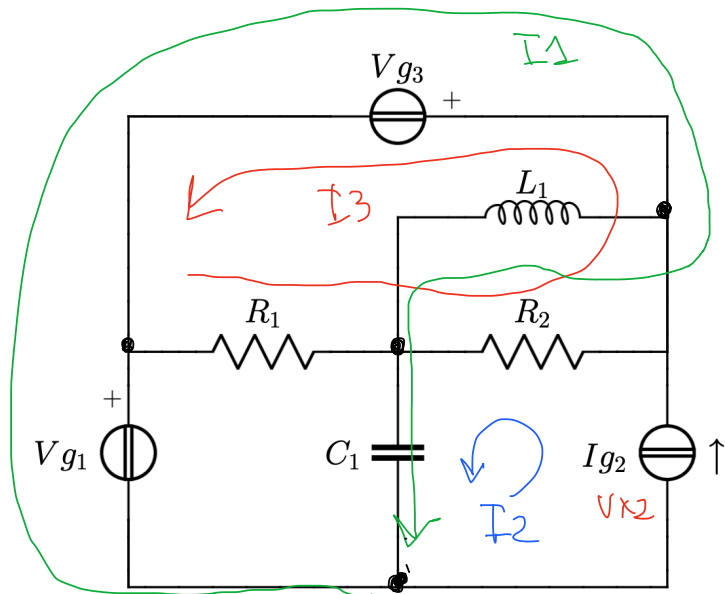


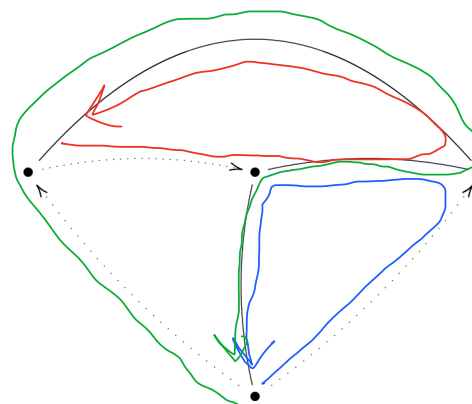
Parallelo Maglie

sabato 2 agosto 2025 15:00

Risolvere il circuito in figura



$$\begin{aligned} V_{g1} &= -2 \\ C_1 &= 1 \\ I_{g2} &= -1 \\ R_1 &= 2 \\ R_2 &= 1 \\ L_1 &= 1 \\ V_{g3} &= 1 + 2j \\ \omega &= 1 \end{aligned}$$



$$Y_a = \frac{1}{3\omega L_1} + \frac{1}{R_2} = \frac{1}{3} + 1 = 1 - j$$

$$Z_a = \frac{1}{Y_a} = \frac{1}{1-j} = \frac{1+j}{2} = \frac{1}{2} + \frac{j}{2}$$

$$\begin{bmatrix} Z_a + \frac{1}{3\omega C_1} & Z_a + \frac{1}{3\omega C_1} & -Z_a \\ Z_a + \frac{1}{3\omega C_1} & Z_a + \frac{1}{3\omega C_1} & -Z_a \\ -Z_a & -Z_a & R_1 + Z_a \end{bmatrix} \begin{pmatrix} I_1 \\ I_2 \\ I_3 \end{pmatrix} = \begin{pmatrix} V_{g3} + V_{g1} \\ V_{x2} \\ -V_{g3} \end{pmatrix}$$

$$\begin{cases} (Z_a + \frac{1}{s\omega C_1}) I_1 + (Z_a + \frac{1}{s\omega C_1}) I_2 - Z_a I_3 = V_{q3} + V_{q1} \\ (Z_a + \frac{1}{s\omega C_1}) I_1 + (Z_a + \frac{1}{s\omega C_1}) I_2 - Z_a I_3 = V_{x2} \\ -Z_a I_1 - Z_a I_2 + (R_1 + Z_a) I_3 = -V_{q3} \\ I_2 = I_{q2} \end{cases}$$

RISOLVIAMO QUESTO SISTEMA.

$$\begin{cases} (\frac{1}{2} + \frac{1}{2}s - 5) I_1 + (\frac{1}{2} + \frac{1}{2}s - 5) I_2 + (-\frac{1}{2} - \frac{1}{2}s) I_3 = 25 - 1 \\ (\frac{1}{2} + \frac{1}{2}s - 5) I_1 + (\frac{1}{2} + \frac{1}{2}s - 5) I_2 + (-\frac{1}{2} - \frac{1}{2}s) I_3 = V_{x2} \\ (-\frac{1}{2} - \frac{1}{2}s) I_1 + (-\frac{1}{2} - \frac{1}{2}s) I_2 + (2 + \frac{1}{2} + \frac{1}{2}s) I_3 = -V_{q3} \\ I_2 = -1 \end{cases}$$

$$\begin{cases} (\frac{1}{2} - \frac{1}{2}s) I_1 + (\frac{1}{2}s - \frac{1}{2}) + (-\frac{1}{2} - \frac{1}{2}s) I_3 = 25 - 1 & (1) \quad \checkmark \\ (\frac{1}{2} - \frac{1}{2}s) I_1 + (\frac{1}{2}s - \frac{1}{2}) + (-\frac{1}{2} - \frac{1}{2}s) I_3 = V_{x2} & (2) \quad \checkmark \\ (-\frac{1}{2} - \frac{1}{2}s) I_1 + (\frac{1}{2} + \frac{1}{2}s) + (\frac{5}{2} + \frac{1}{2}s) I_3 = -25 - 1 & (3) \quad \checkmark \end{cases}$$

DALLA PRIMA RICOVO I_1

LO SOSTITUISCO NELLA (3)

$$\left(\frac{1}{2}-\frac{1}{2}s\right)I_1 + \left(\frac{1}{2}s-\frac{1}{2}\right) + \left(-\frac{1}{2}-\frac{1}{2}s\right)I_3 = 2s-1$$

$$\left(\frac{1-s}{2}\right)I_1 = 2s-1 + \frac{1}{2} - \frac{1}{2}s + \frac{1}{2}I_3 + \frac{1}{2}sI_3$$

$$(1-s)I_1 = 4s-2 + 1-s + I_3 + sI_3$$

$$I_1 = \frac{4s-2+1-s+I_3+sI_3}{1-s}$$

$$I_1 = \frac{3s-1+(s+1)I_3}{1-s}$$

$$\rightarrow \frac{3s-1+(s+1)I_3}{1-s} \cdot \frac{1+s}{1+s}$$

$$= \frac{3s-3-1-s+5I_3-I_3+I_3+sI_3}{2} = \frac{2s-4+2sI_3}{2} = s-2+sI_3$$

$$I_1 = s-2+s(-s-1) = s-2+1-s = -1$$

ORA RICAVO U_{X2} dalla (2)

$$\left(1-\frac{1}{2}s\right)I_1 + \left(\frac{1}{2}s-\frac{1}{2}\right) + \left(-\frac{1}{2}-\frac{1}{2}s\right)I_3 = U_{X2}$$

$$\left(-\frac{1}{2}-\frac{1}{2}s\right)(s-2+sI_3) + \frac{s}{2}I_3 + \frac{1}{2}sI_3 = -2s-1-\frac{s}{2}-\frac{s}{2}$$

$$-\frac{1}{2}s + 1 - \frac{1}{2}sI_3 + \frac{1}{2}s + s + \frac{1}{2}I_3 + \frac{s}{2}I_3 + \frac{1}{2}sI_3 = -\frac{5}{2}s - \frac{3}{2}$$

$$-\frac{1}{2}sI_3 + \frac{1}{2}I_3 + \frac{s}{2}I_3 + \frac{1}{2}sI_3 =$$

$$= -\frac{5}{2}s - \frac{3}{2} - 1 + \frac{1}{2}s - \frac{1}{2}s$$

$$= 3I_3 = -3s-3$$

$$= I_3 = -s-1$$

$$\frac{1}{2}5 - \frac{1}{2} + \frac{1}{2}5 - \frac{1}{2} + \left(-\frac{1}{2} - \frac{1}{2}5\right)(-3-1) = V \times 2$$

$$5 - 1 + \frac{1}{2}5 + \frac{1}{2} - \frac{1}{2} + \frac{1}{2}5 = V \times 2$$

$$5 - 1 + 5 = V \times 2$$

$$V \times 2 = 25 - 1$$

Riassumendo:

$$\begin{cases} I_1 = -1 \checkmark \\ I_2 = -1 \checkmark \\ I_3 = -3 - 1 \checkmark \\ V \times 2 = 25 - 1 \checkmark \end{cases}$$

BILANCIO DI POTENZA!

$$I_{R1} = I_3 = -3 - 1$$

$$I_{R2} + I_{L1} = I_3 - I_1 - I_2 = -3 - 1 + 1 + 1 = 1 - 3$$

$$I_{C1} = -I_1 - I_2 = 1 + 1 = 2$$

$$I_{R2} = -1 \text{ (nota)}$$

$$V_{R1} = R_1 \cdot I_{R1} = 2 \cdot (-3 - 1) = -25 - 2$$

$$V_{R2} = V_{L1} = Z_{\omega} \cdot (I_{R2} + I_{L1}) = \left(\frac{1}{2} + \frac{1}{2}5\right)(1 - 3) = 1$$

$$V_{C1} = \frac{1}{S_{WC1}} \cdot I_{C1} = \frac{1}{5} \cdot 2 = -25$$

$$V_{R1} = -2 \text{ (nota)}$$

$$I_{g1} = I_1 = -1$$

$$I_{g3} = I_1 - I_3 = -1 + 5 + 1 = 5$$

$$R_2 = \frac{V_{R2}}{I_{R2}} = \frac{1}{1} = 1$$

$$L_1 = \frac{V_{L1}}{I_{L1}} = \frac{1}{-5} = -\frac{1}{5}$$

$$V_{g3} = 1 + 25 (V_{oTb})$$

$$V_{g2} = V_{x2} = 25 - 1$$

$$P_{C V_{g1}} = \frac{1}{2} V_{g1} I_{g1}^* = \frac{1}{2} (-2) (-1) = 1$$

$$P_{C I_{g2}} = \frac{1}{2} V_{g2} I_{g2}^* = \frac{1}{2} (25 - 1) (-1) = -\left(5 - \frac{1}{2}\right) = -\frac{1}{2} - 5$$

$$P_{C I_{g3}} = \frac{1}{2} V_{g3} I_{g3}^* = \frac{1}{2} (1 + 25) (-5) = -\left(\frac{1}{2} + 5\right) (-5) = -\frac{1}{2} (-5 + 1)$$

$$1 + \frac{1}{2} + 1 = 2 + \frac{1}{2}$$

$$\frac{5}{2} = \frac{5}{2}$$

$$P_{AR1} = \frac{1}{2} R_1 |I_{R1}|^2 = \frac{1}{2} 2 \cdot 2 = 2$$

$$P_{AR2} = \frac{1}{2} R_2 |I_{R2}|^2 = \frac{1}{2} 1 \cdot 1 = \frac{1}{2}$$

$$Q_{L1} = \frac{1}{2} \omega L_1 |I_{L1}|^2 = \frac{1}{2} 1 = \frac{1}{2}$$

$$Q_{C1} = -\frac{1}{2} \omega C_1 |V_{C1}|^2 = -\frac{1}{2} 1 \cdot 4 = -2$$

$$P_{R_{g1}} = \operatorname{Im}(P_{C V_{g1}}) = 0$$

$$P_{R_{g2}} = \operatorname{Im}(P_{C I_{g2}}) = -1$$

$$P_{R_{g3}} = \operatorname{Im}(P_{C I_{g3}}) = -\frac{1}{2}$$

$$\frac{1}{2} - 2 = -1 - \frac{1}{2}$$

$$-\frac{3}{2} = -\frac{3}{2}$$

